Mack's chain-ladder model in Lean 4: blueprint

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Contents

Chapter 1

Introduction

This blueprint tracks the formalization of Mack's distribution-free chain-ladder model [1] in Lean 4 on top of mathlib. Each statement below carries the name of its Lean declaration; green nodes in the dependency graph are proved, the others are open work listed in the roadmap. Statements are given in the paper's own terms, and where a published formula is an estimator or an approximation rather than an identity, the text says so.

A run-off triangle with n accident years and n development years is a function $C : \mathbb{N} \to \mathbb{N} \to \mathbb{R}$; $C_{i,k}$ is the cumulative claims of accident year i at development year k, observed when $i+k \leq n-1$. The latest observed entry of accident year i is $C_{i,\,n-1-i}$.

Chapter 2

The chain-ladder estimators

2.1 Definitions

Definition 2.1 (Contributors)**.** The accident years contributing to the development factor $k \to k+1$ are $i = 0, \dots, n-k-2$, those with $C_{i,k+1}$ observed.

Definition 2.2 (Column sums)**.** $S_k = \sum_{i=0}^{n-k-2} C_{i,k}$ and $T_k = \sum_{i=0}^{n-k-2} C_{i,k+1}$.

Definition 2.3 (Development factor)**.** $\hat{f}_k = T_k / S_k$. (In Lean, division by zero is zero.)

Definition 2.4 (Individual factor)**.** $F_{i,k} = C_{i,k+1} / C_{i,k}$.

Definition 2.5 (Mack's variance estimator)**.** $\hat{\sigma}_k^2 = \frac{1}{n-k-2} \sum_{i=0}^{n-k-2} C_{i,k} \, (F_{i,k} - \hat{f}_k)^2$. The value for $k = n-2$ (a single contributor) is an extrapolation convention in [1], not an estimator; no theorem here uses it.

Definition 2.6 (Projection)**.** $\hat{C}_{i,k} = C_{i,n-1-i} \prod_{j=n-1-i}^{k-1} \hat{f}_j$ for $k \geq n-1-i$.

Definition 2.7 (Ultimate and reserve)**.** $\hat{C}_{i,n-1}$ is the chain-ladder ultimate and $\hat{R}_i = \hat{C}_{i,n-1} - C_{i,n-1-i}$ the reserve.

Definition 2.8 (Mack's MSEP estimator)**.** Mack (1993), Theorem 3:

$$\widehat{\mathrm{msep}}(\hat{R}_i) = \hat{C}_{i,n-1}^2 \sum_{k=n-1-i}^{n-2} \frac{\hat{\sigma}_k^2}{\hat{f}_k^2} \Big(\frac{1}{\hat{C}_{i,k}} + \frac{1}{S_k} \Big).$$

2.2 Deterministic identities

Lemma 2.9 (Weighted-average form)**.** *If $C_{i,k} \neq 0$ for every contributor i, then $\hat{f}_k = \sum_i \frac{C_{i,k}}{S_k} F_{i,k}$.*

Proof. Termwise: $\frac{C_{i,k}}{S_k} \cdot \frac{C_{i,k+1}}{C_{i,k}} = \frac{C_{i,k+1}}{S_k}$. □

Lemma 2.10. *If $C_{i,k} \neq 0$ for every contributor, $T_k = \sum_i C_{i,k} F_{i,k}$.*

Proof. Cancel the denominator termwise. □

Lemma 2.11 (Sum-of-squares decomposition)**.** *For any centre f, with $C_{i,k} \neq 0$ for every contributor, $\sum_i C_{i,k} (F_{i,k} - f)^2 = \sum_i C_{i,k} F_{i,k}^2 - 2 f T_k + f^2 S_k$.*

Proof. Expand the square and use Lemma 2.10. □

Lemma 2.12 (Collapse at the chain-ladder centre)**.** *If moreover $S_k \neq 0$, then $\sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2 = \sum_i C_{i,k}F_{i,k}^2 - S_k\hat{f}_k^2$. This is the identity behind the $n-k-2$ degrees of freedom in $\hat{\sigma}_k^2$.*

Proof. Put $f = \hat{f}_k = T_k/S_k$ in Lemma 2.11. □

Lemma 2.13 (Projection step)**.** *For $k \geq n-1-i$, $\hat{C}_{i,k+1} = \hat{C}_{i,k}\,\hat{f}_k$.*

Proof. Split off the top factor of the product. □

Lemma 2.14. $\hat{C}_{i,n-1-i} = C_{i,n-1-i}$.

Proof. Empty product. □

Lemma 2.15. *The oldest accident year is fully developed: $\hat{R}_0 = 0$.*

Proof. Empty product again. □

Chapter 3

Mack 1999 equals Mack 1993

Definition 3.1 (Mack's 1999 recursion)**.** For $\alpha = 1$ and unit weights [2], starting from $\mathrm{se}^2 = 0$ on the latest diagonal $d = n - 1 - i$ and stepping m times,

$$\mathrm{se}^2_{m+1} = \hat{C}^2_{i,d+m}\Big(\frac{\hat{\sigma}^2_{d+m}}{\hat{C}_{i,d+m}} + \frac{\hat{\sigma}^2_{d+m}}{S_{d+m}}\Big) + \mathrm{se}^2_m\,\hat{f}^2_{d+m}.$$

Definition 3.2. The summand of the closed form: $\mathrm{term}_{i,k} = \frac{\hat{\sigma}^2_k}{\hat{f}^2_k}\big(\frac{1}{\hat{C}_{i,k}} + \frac{1}{S_k}\big)$.

Lemma 3.3. $\widehat{\mathrm{msep}}(\hat{R}_i) = \hat{C}^2_{i,n-1}\sum_{k=n-1-i}^{n-2}\mathrm{term}_{i,k}$.

Proof. By definition. □

Lemma 3.4 (Truncated closed form)**.** *If $\hat{f}_k \neq 0$ for $d \le k < d{+}m$, then* $\mathrm{se}^2_m = \hat{C}^2_{i,d+m}\sum_{k=d}^{d+m-1}\mathrm{term}_{i,k}$.

Proof. Induction on m using Lemma 2.13 and $\hat{f}^2_{d+m}/\hat{f}^2_{d+m} = 1$. □

Theorem 3.5 (Mack 1999 = Mack 1993)**.** *For $i \le n-1$ and $\hat{f}_k \neq 0$ along the row, i steps of the 1999 recursion return exactly Mack's 1993 estimator:* $\mathrm{se}^2_i = \widehat{\mathrm{msep}}(\hat{R}_i)$.

Proof. Lemma 3.4 with $m = i$, since $d + i = n - 1$. □

Chapter 4

Bornhuetter-Ferguson on the chain-ladder pattern

Definition 4.1 (Cumulative development factor, BF reserve and ultimate)**.** $\mathrm{cdf}_i = \prod_{k=d}^{n-2} \hat{f}_k$; for an a priori ultimate U, $R_i^{BF} = U\,(1 - 1/\mathrm{cdf}_i)$ and $\hat{C}_{i,n-1}^{BF} = C_{i,d} + R_i^{BF}$.

Lemma 4.2 (Linearity)**.** *$R_i^{BF}(aU) = a\,R_i^{BF}(U)$ and $R_i^{BF}(U+V) = R_i^{BF}(U) + R_i^{BF}(V)$.*

Proof. Ring identities. ☐

Theorem 4.3 (BF with the chain-ladder prior is chain ladder)**.** *If $\mathrm{cdf}_i \neq 0$ and $U = \hat{C}_{i,n-1}$, then $\hat{C}_{i,n-1}^{BF} = \hat{C}_{i,n-1}$ and $R_i^{BF} = \hat{R}_i$.*

Proof. $\hat{C}_{i,n-1} = C_{i,d}\,\mathrm{cdf}_i$, so $U(1 - 1/\mathrm{cdf}_i) = C_{i,d}\mathrm{cdf}_i - C_{i,d}$. ☐

Lemma 4.4 (The BF reserve is a fraction of the prior)**.** *If every development factor along the row is at least* 1*, then $\mathrm{cdf}_i \geq 1$ and, for $U \geq 0$, $0 \leq R_i^{BF}(U) \leq U$.*

Proof. A product of numbers at least 1 is at least 1; then $0 \leq 1 - 1/\mathrm{cdf}_i \leq 1$. ☐

Chapter 5

The stochastic layer

5.1 Model

Definition 5.1 (Random triangle)**.** A random run-off triangle on a measurable space Ω is a family of random variables $C_{i,k} : \Omega \to \mathbb{R}$ together with a filtration $\mathcal{D}_k$ such that $C_{i,j}$ is $\mathcal{D}_k$-measurable whenever $j \leq k$ ("everything observed up to development year k").

Definition 5.2 (Assumption (M1))**.** $E[C_{i,k+1} \mid \mathcal{D}_k] = f_k\, C_{i,k}$ almost surely, for all i and k. This is Mack's first assumption in the form conditioned on $\mathcal{D}_k$; Mack states it conditioned on the row's own history and obtains this form from independence across accident years (M2). That passage is Theorem 5.30.

Definition 5.3 (Estimators as random variables)**.** $S_k(\omega)$ and $\hat{f}_k(\omega)$ are the deterministic estimators applied to the triangle observed at ω.

Lemma 5.4. *S_k is $\mathcal{D}_k$-measurable.*

Proof. A finite sum of $\mathcal{D}_k$-measurable functions. □

5.2 Mack's Theorem 2

Theorem 5.5 (Conditional unbiasedness of the development factor)**.** *Assume (M1), that $S_k \neq 0$ almost surely, that each $C_{i,k+1}$ and $\hat{f}_k$ are integrable. Then $E[\hat{f}_k \mid \mathcal{D}_k] = f_k$ almost surely.*

Proof. $\hat{f}_k = S_k^{-1} \sum_i C_{i,k+1}$. The factor S_k^{-1} is $\mathcal{D}_k$-measurable and comes out of the conditional expectation; by (M1) the conditional expectation of the sum is $f_k \sum_i C_{i,k} = f_k S_k$; cancel S_k where it is nonzero. □

Lemma 5.6. *$\hat{f}_j$ is $\mathcal{D}_k$-measurable whenever $j + 1 \leq k$.*

Proof. S_j is $\mathcal{D}_j \subseteq \mathcal{D}_k$-measurable and each $C_{i,j+1}$ is $\mathcal{D}_k$-measurable. □

Theorem 5.7 (Uncorrelated development factors, conditional form)**.** *Under (M1), on a finite measure space, for $j < k$ (with the integrability and $S \neq 0$ hypotheses of Theorem 5.5 for both columns and $\hat{f}_j \hat{f}_k$ integrable), $E[\hat{f}_j \hat{f}_k \mid \mathcal{D}_j] = f_j f_k$ almost surely.*

Proof. Condition on $\mathcal{D}_k$ first: $\hat{f}_j$ is $\mathcal{D}_k$-measurable and comes out, and $E[\hat{f}_k \mid \mathcal{D}_k] = f_k$ by Theorem 5.5. Then the tower property down to $\mathcal{D}_j$ and Theorem 5.5 again for column j. □

Theorem 5.8 (Mack 1993, Theorem 2). *On a probability space, under the same hypotheses, $E[\hat{f}_k] = f_k$ and $E[\hat{f}_j \hat{f}_k] = f_j f_k$ for $j < k$: the chain-ladder factors are unbiased and uncorrelated, which is what makes $\prod_k \hat{f}_k$ an unbiased estimator of $\prod_k f_k$.*

Proof. Integrate the conditional statements. □

5.3 Mack's Theorem 1

Definition 5.9 (Chain-ladder projection as a random variable). $\hat{C}_{i,d+m} = C_{i,d} \prod_{k=d}^{d+m-1} \hat{f}_k$ with $d = n - 1 - i$.

Lemma 5.10. *$\hat{C}_{i,d+m}$ is $\mathcal{D}_{d+m}$-measurable.*

Proof. Induction on m; each new factor $\hat{f}_{d+m}$ is $\mathcal{D}_{d+m+1}$-measurable. □

Lemma 5.11 (Iterating (M1)). *For $d \le k$, $E[C_{i,k+1} \mid \mathcal{D}_d] = f_k\, E[C_{i,k} \mid \mathcal{D}_d]$, hence $E[C_{i,d+m} \mid \mathcal{D}_d] = C_{i,d} \prod_{k=d}^{d+m-1} f_k$.*

Proof. Tower property through $\mathcal{D}_k$, then (M1); induction on m. □

Theorem 5.12 (Mack 1993, Theorem 1). *Under (M1), with the partial products integrable and the column sums a.s. nonzero, $E[\hat{C}_{i,d+m} \mid \mathcal{D}_d] = C_{i,d} \prod_{k=d}^{d+m-1} f_k$, and therefore $E[\hat{C}_{i,n-1} \mid \mathcal{D}_d] = E[C_{i,n-1} \mid \mathcal{D}_d]$: the chain-ladder ultimate is a conditionally unbiased estimator of the true ultimate claims.*

Proof. Condition on $\mathcal{D}_{d+m}$: the partial product comes out and the last factor has conditional expectation f_{d+m} (Theorem 5.5); tower down to $\mathcal{D}_d$ and induct. Compare with Lemma 5.11. □

5.4 The variance assumptions and the estimation variance of $\hat{f}_k$

Definition 5.13 (Residuals, (M3) and (M2')). With $\varepsilon_{i,k} = C_{i,k+1} - f_k C_{i,k}$: (M3) $E[\varepsilon_{i,k}^2 \mid \mathcal{D}_k] = \sigma_k^2 C_{i,k}$, and (M2') $E[\varepsilon_{i,k}\varepsilon_{j,k} \mid \mathcal{D}_k] = 0$ for $i \neq j$, the conditional uncorrelatedness across accident years that independence of accident years supplies.

Lemma 5.14. *On $\{S_k \neq 0\}$, $\hat{f}_k - f_k = S_k^{-1} \sum_i \varepsilon_{i,k}$ and $(\hat{f}_k - f_k)^2 = S_k^{-2} \sum_i \sum_j \varepsilon_{i,k}\varepsilon_{j,k}$.*

Proof. $T_k - f_k S_k = \sum_i \varepsilon_{i,k}$; expand the square of the sum. □

Theorem 5.15 (Estimation variance of the development factor). *Under (M3) and (M2'), with the residual products integrable, on $\{S_k \neq 0\}$, $E[(\hat{f}_k - f_k)^2 \mid \mathcal{D}_k] = \sigma_k^2 / S_k$ (Mack 1993, proof of Theorem 3; Mack 1999, s.e.$(\hat{f}_k)^2 = \hat{\sigma}_k^2 / S_k$).*

Proof. S_k^{-2} is $\mathcal{D}_k$-measurable and comes out; conditional expectation is linear over the double sum; cross terms vanish by (M2'); diagonal terms give $\sigma_k^2 \sum_i C_{i,k} = \sigma_k^2 S_k$. □

5.5 Unbiasedness of the variance estimator

Lemma 5.16 (Sum of squares around $\hat{f}_k$ via residuals)**.** *With all contributing $C_{i,k} \neq 0$ and $S_k \neq 0$, for any f, $\sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2 = \sum_i \varepsilon_i^2/C_{i,k} - S_k(\hat{f}_k - f)^2$ where $\varepsilon_i = C_{i,k+1} - fC_{i,k}$.*

Proof. Two instances of Lemma 2.11 (centres $\hat{f}_k$ and f) and $T_k = \hat{f}_k S_k$. □

Definition 5.17. $\hat{\sigma}_k^2$ and the weighted sum of squares $W_k = \sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2$ as random variables; $\hat{\sigma}_k^2 = W_k/(n-k-2)$.

Theorem 5.18 (Unbiasedness of the variance estimator)**.** *Under (M3) and (M2'), with all contributing $C_{i,k}$ and S_k a.s. nonzero and the stated integrability, $E[W_k \mid \mathcal{D}_k] = (n-k-2)\,\sigma_k^2$ for $k+2 \leq n$, and hence $E[\hat{\sigma}_k^2 \mid \mathcal{D}_k] = \sigma_k^2$ for $k+3 \leq n$.*

Proof. By Lemma 5.16, $W_k = \sum_i \varepsilon_i^2/C_{i,k} - S_k(\hat{f}_k - f_k)^2$. Each $C_{i,k}^{-1}$ is $\mathcal{D}_k$-measurable, so (M3) gives σ_k^2 per term, $n-k-1$ terms in all; S_k is $\mathcal{D}_k$-measurable and Theorem 5.15 gives σ_k^2 for the subtracted term. □

5.6 Process variance and the exact MSEP (Theorem 3)

Lemma 5.19 (Conditional second moment)**.** *Under (M1) and (M3), $E[\varepsilon_{i,k} \mid \mathcal{D}_k] = 0$, $E[C_{i,k+1}^2 \mid \mathcal{D}_k] = f_k^2 C_{i,k}^2 + \sigma_k^2 C_{i,k}$, and for $d \leq k$, $E[C_{i,k+1}^2 \mid \mathcal{D}_d] = f_k^2 E[C_{i,k}^2 \mid \mathcal{D}_d] + \sigma_k^2 E[C_{i,k} \mid \mathcal{D}_d]$.*

Proof. Expand $C_{i,k+1} = f_k C_{i,k} + \varepsilon_{i,k}$; the cross term vanishes; tower property. □

Definition 5.20 (Process variance)**.** $V_0 = 0$, $V_{m+1} = f_{d+m}^2 V_m + \sigma_{d+m}^2\, c \prod_{l<m} f_{d+l}$; in closed form $V_m = \sum_{j<m} \left(\prod_{l=j+1}^{m-1} f_{d+l}\right)^2 \sigma_{d+j}^2\, c \prod_{l<j} f_{d+l}$.

Theorem 5.21 (Process variance along a row)**.** *Under (M1) and (M3), $E[C_{i,d+m}^2 \mid \mathcal{D}_d] - E[C_{i,d+m} \mid \mathcal{D}_d]^2 = V_m$ with $c = C_{i,d}$.*

Proof. Induction on m with Lemma 5.19 and Lemma 5.11. □

Lemma 5.22 (Conditional MSEP decomposition)**.** *For a $\mathcal{D}$-measurable predictor P of Y, $E[(P - Y)^2 \mid \mathcal{D}] = \left(E[Y^2 \mid \mathcal{D}] - E[Y \mid \mathcal{D}]^2\right) + (P - E[Y \mid \mathcal{D}])^2$.*

Proof. Expand the square; P comes out of the conditional expectation. □

Theorem 5.23 (Mack's Theorem 3, exact form)**.** *Let $\mathcal{D}$ be the observed data. Assume $\hat{C}_{i,d+m}$ is $\mathcal{D}$-measurable and that $\mathcal{D}$ adds nothing to $\mathcal{D}_d$ about row i's future (the conditional mean and second moment of $C_{i,d+m}$ given $\mathcal{D}$ agree with those given $\mathcal{D}_d$; this is what independence across accident years supplies). Then under (M1) and (M3)*

$$E\left[(\hat{C}_{i,d+m} - C_{i,d+m})^2 \mid \mathcal{D}\right] = V_m + \left(\hat{C}_{i,d+m} - C_{i,d} \prod_{l<m} f_{d+l}\right)^2.$$

Proof. Lemma 5.22 with $P = \hat{C}_{i,d+m}$, then Theorem 5.21 and Lemma 5.11. □

Definition 5.24 (Mack's estimators of the two terms)**.** $\widehat{V}$ is V_m with $\hat{f}, \hat{\sigma}^2$ plugged in. The estimation-error term $\hat{C}_{i,n-1}^2 \sum_k \hat{\sigma}_k^2/(\hat{f}_k^2 S_k)$ is obtained from $(\hat{C} - M)^2$ by Mack's conditional-resampling step: replace $(\hat{f}_k - f_k)^2$ by σ_k^2/S_k (Theorem 5.15), drop the cross terms (Theorem 5.7), and plug in $\hat{f}, \hat{\sigma}^2$. This is a definition, not a theorem: it is the approximation in Mack (1993), and the exact conditional MSEP above is the object it approximates (Buchwalder, Bühlmann, Merz and Wüthrich 2006 estimate the same object differently).

5.7 From independence across accident years to the $\mathcal{D}_k$ form

Mack states (M1) and (M3) conditioned on one accident year's own history, and assumes separately that the accident years are independent (M2). Every theorem above conditions on $\mathcal{D}_k$ instead. This section proves the passage.

Definition 5.25 (Row σ-algebras)**.** $\sigma(C_{i,0}, \dots, C_{i,k})$, the history of accident year i up to development year k; $\sigma(C_{i,j} : j)$, the whole of accident year i; and the joins of these over the other accident years $i' \neq i$.

Definition 5.26 ((M2) and the row-conditioned assumptions)**.** (M2) is `RowsIndep`: the σ-algebras of the individual accident years are independent. `RowsGenerateD` says that $\mathcal{D}_k$ is what its name means, the join of the rows' histories up to development year k. (M1row) and (M3row) are Mack's assumptions conditioned on $\sigma(C_{i,0}, \dots, C_{i,k})$ rather than on $\mathcal{D}_k$.

Lemma 5.27 (Rectangles)**.** *If m_1' and m_2 are independent, F is m_1'-measurable and integrable, $a \in m_1'$ and $b \in m_2$, then $\int_{a \cap b} F = \mu(b) \int_a F$.*

Proof. Apply mathlib's `condExp_indep_eq` to the indicator $1_a F$, which is m_1'-measurable, and integrate the resulting constant over b. □

Theorem 5.28 (Conditioning on an independent enlargement changes nothing)**.** *Let $m_1 \leq m_1'$, let f be m_1'-measurable and integrable, and let m_2 be independent of m_1'. Then $E[f \mid m_1 \vee m_2] = E[f \mid m_1]$ almost surely.*

Proof. The sets $a \cap b$ with $a \in m_1$, $b \in m_2$ form a π-system generating $m_1 \vee m_2$. On such a set both $\int f$ and $\int E[f \mid m_1]$ factor by Lemma 5.27 into $\mu(b)$ times the integral over a, and those agree by the defining property of $E[f \mid m_1]$. The class of sets on which the two set integrals agree is closed under complements and countable disjoint unions, so it is all of $m_1 \vee m_2$; conclude by the uniqueness of the conditional expectation. Mathlib has the case $m_1 = \perp$ only. □

Lemma 5.29 (One row against the others)**.** *Under (M2), accident year i is independent of all the other accident years taken together, and so of their observed histories. Under `RowsGenerateD`, $\mathcal{D}_k$ is the join of row i's history up to k with the other rows' histories up to k.*

Proof. `indep_iSup_of_disjoint` for $\{i\}$ against its complement, then monotonicity of independence; the splitting of $\mathcal{D}_k$ is `iSup_split_single`. □

Theorem 5.30 ((M1) and (M3) in the $\mathcal{D}_k$ form)**.** *Assume (M2), that $\mathcal{D}_k$ is generated by the rows' histories, and the integrability of the entries. Then Mack's row-conditioned (M1row) implies (M1), and (M3row) implies (M3).*

Proof. Split $\mathcal{D}_k$ by Lemma 5.29, apply Theorem 5.28 with m_1 the row's history up to k, m_1' the whole row and m_2 the other rows, and finish with the row-conditioned assumption. □

Theorem 5.31 ((M2') is a consequence, not an assumption)**.** *Under (M2), `RowsGenerateD` and (M1row), the residuals of two different accident years are conditionally uncorrelated given $\mathcal{D}_k$: assumption (M2'), which the variance theorems take as a hypothesis, follows from independence across accident years.*

Proof. Condition on $\mathcal{D}_k$ together with the whole of accident year j: row j's residual comes out of the conditional expectation, and what is left is row i's residual conditioned on a σ-algebra that adds nothing about row i, so it equals $E[\varepsilon_{i,k} \mid \sigma(C_{i,0}, \dots, C_{i,k})] = 0$ by Theorem 5.28 and (M1row). The tower property brings the identity back to $\mathcal{D}_k$. □

Chapter 6

The variant catalogue: Mack versus conditional resampling

Write $a_k = \hat{\sigma}_k^2/(\hat{f}_k^2 S_k)$ for the relative estimation variance of development factor k along the row. Mack's estimation-error term is $\hat{C}_{i,n-1}^2 \sum_k a_k$; the conditional-resampling term of Buchwalder, Bühlmann, Merz and Wüthrich (2006), which Murphy (1994) also reaches, is $\hat{C}_{i,n-1}^2 \big(\prod_k (1 + a_k) - 1 \big)$.

Definition 6.1. a_k and the conditional-resampling estimator $\hat{C}_{i,n-1}^2 (\prod_k (1 + a_k) - 1)$.

Lemma 6.2 (Product minus sum)**.** *For $a_k \geq 0$: $1 + \sum a_k \leq \prod (1 + a_k) \leq \exp(\sum a_k)$; for two terms $(1 + a)(1 + b) - 1 - (a + b) = ab$; and $\prod (1 + a_k) - 1 - \sum a_k = 0$ when at most one a_k is nonzero.*

Proof. Induction over the index set for the Weierstrass inequality; $1 + x \leq e^x$ termwise for the exponential bound. □

Theorem 6.3 (Catalogue row: Mack is the first-order part of BBMW)**.** *With $a_k \geq 0$ along the row: (i) Mack $\leq$ BBMW; (ii) the difference is exactly $\hat{C}_{i,n-1}^2 \big(\prod_k (1 + a_k) - 1 - \sum_k a_k \big)$; (iii) the two coincide for the second-latest accident year, which has a single development factor; (iv) the difference is at most $\hat{C}_{i,n-1}^2 \big(e^{\sum a_k} - 1 - \sum a_k \big)$, second order in the relative estimation variances.*

Proof. Lemma 6.2 scaled by $\hat{C}_{i,n-1}^2$. □

Theorem 6.4 (Strict counterexample)**.** *There is a run-off triangle (four accident years, four development years, with unequal individual factors in the first two columns) and an accident year on which Mack's estimation-error term is strictly smaller than the conditional-resampling term: the two are different estimators, not two presentations of one. On the exhibited triangle $a_0 = 1/297$, $a_1 = 1/2809$, and the remainder is $\hat{C}^2 a_0 a_1 > 0$. Every number is checked by* `norm_num` *from the definitions.*

Proof. Evaluate the definitions on the triangle; apply Theorem 6.3(ii). □

6.1 The total reserve: Mack's Corollary

Definition 6.5 (Total-reserve estimators)**.** Mack's Corollary: $\widehat{\mathrm{msep}}(\hat{R}) = \sum_i \widehat{\mathrm{msep}}_i + \sum_i \hat{C}_i \big(\sum_{j>i} \hat{C}_j\big)\, 2 \sum_{k \in \mathrm{row}\, i} a_k$, the cross terms running over the older year's remaining factors. The conditional-resampling aggregate replaces $2\sum_k a_k$ by $2(\prod_k (1+a_k) - 1)$ (Buchwalder et al. 2006, Section 4.3).

Theorem 6.6 (Aggregated catalogue row)**.** *The aggregated estimation-error terms differ by* $\sum_i \big(\hat{C}_i^2 + 2\hat{C}_i \sum_{j>i} \hat{C}_j\big)\big(\prod_{k\in\mathrm{row}\, i}(1+a_k) - 1 - \sum_{k\in \mathrm{row}\, i} a_k\big)$*, which is nonnegative when the ultimates and the* a_k *are: Mack's total is at most the resampling total, by the same row-wise second-order remainder as in the single-year case.*

Proof. Termwise algebra and Lemma 6.2. □

Chapter 7

Non-vacuity: a nondegenerate model

Theorem 7.1 (A nondegenerate Mack model exists)**.** *There is a probability space, a random triangle with three accident years, and constants f_k, σ_k^2 with $\sigma_0^2 > 0$, satisfying (M1), (M3) and (M2'), in which the first development step is genuinely random. The model: eight equally likely outcomes, an independent ± 1 shock per accident year, $C_{i,0} = c$ and $C_{i,k} = cf^k(1 + s\,\xi_i)$ for $k \geq 1$, with $\mathcal{D}_0$ trivial and $\mathcal{D}_k$ everything for $k \geq 1$.*

Proof. Conditional expectations given the trivial σ-algebra are eight-term averages; the shocks sum to zero, have unit square, and are orthogonal. □

Theorem 7.2 (Headline theorems instantiated)**.** *On that model, $E[\hat{f}_0 \mid \mathcal{D}_0] = 2$ although $\hat{f}_0$ is random; $E[\hat{C}_{2,2} \mid \mathcal{D}_0] = E[C_{2,2} \mid \mathcal{D}_0]$; $E[(\hat{f}_0 - 2)^2 \mid \mathcal{D}_0] = \sigma_0^2/S_0$; and $E[\hat{\sigma}_0^2 \mid \mathcal{D}_0] = \sigma_0^2 = 4$.*

Proof. Direct instantiation of the general theorems; every hypothesis is discharged. □

Chapter 8

Reference computation

Mack's 1993 example (the Taylor and Ashe 1983 triangle) is reproduced to every printed digit (nine reserves, the total 18,681 thousand, all standard-error percentages) by a dependency-free script implementing the definitions above in floating point. The Lean definitions are over $\mathbb{R}$ and noncomputable; the script is the reference computation, not the Lean code.

Bibliography

[1] T. Mack, Distribution-free calculation of the standard error of chain ladder reserve estimates, ASTIN Bulletin 23 (1993) 213-225.

[2] T. Mack, The standard error of chain ladder reserve estimates: recursive calculation and inclusion of a tail factor, ASTIN Bulletin 29 (1999) 361-366.